Fiscal Year:	FY 2025	**Task Last Updated:**	FY 08/17/2024
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Project Title:	Multiscale Computational Modeling of Dusty Plasmas Near Space Surfaces		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Physical Sciences Informatics (PSI)	**Solicitation / Funding Source:**	2020 Physical Sciences NNH20ZDA014N: Use of the NASA Physical Sciences Informatics System – Appendix G
Start Date:	10/18/2021	**End Date:**	10/17/2023
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
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Grant/Contract No.:	JPL Task - RSA# 1670489		
Performance Goal No.:			
Performance Goal Text:			
	The goals of the proposed two-year research are to: develop and validate with Physical Sciences Informatics (PSI) data a state-of-the-art computational capability for modeling the release and transport of dust particles near floating and biased surfaces in space plasmas and use the multiscale capability and explore release and transport phenomena for spherical and non-spherical dust particles, under plasma and surface conditions beyond those addressed in the PSI experiment. The first objective is to develop a state-of-the-art dusty plasma computational modeling capability from the grain-scale to multiscale. The microscopic (grain-scale) Dusty Parallel Immersed Finite Element Particle-in-Cell (D-IFEPIC) model will be developed on the existing Parallel Immersed Finite Element Particle-in-Cell (PIFEPIC), a parallelized Particle-in-Cell (PIC) platform. The D-IFEPIC model will include spherical and non-spherical grains and will account for both surface and in-depth charging of each grain. D-IFEPIC will resolve the geometrical and material properties		

Task Description:	(permittivity) of each grain and model the unsteady and stochastic charging of grains via a deposition process. The second objective is to simulate the PSI experiment, validate the D-IFEPIC and SPIS codes, and quantify the plasma conditions under which the fluctuating charge on grains near a floating surface reaches a threshold such that the electric field force overcomes the adhesive van der Waals force. The two codes will be used with direct inputs from the PSI data and address comprehensively the multiscale interactions from grain-scale to system-scale observed in the PSI experiment. The third objective is to use D-IFEPIC and investigate the parametric dependence of dust charging and release near floating and biased surfaces for a broad range of plasma and surface conditions beyond those found in the PSI experiment. These simulations will use spherical grains from 0.1-100 um and irregularly shaped grains (D-IFEPIC only), floating and biased surfaces in fully ionized plasmas found in low-Earth orbit (LEO), geosynchronous orbit (GEO), solar, and planetary environments. The results will document the dependence of charge and release time on these parameters and relative role of forces on grains. The comparisons will demonstrate the difference between the capacitance and the surface/in-depth charging models. The D-IFEPIC simulations with non-spherical grains will also demonstrate for the first time the impact of grain shape on sheath electrodynamics. The proposed research directly addresses the scope of the solicitation: provides testing of hypothesis of dust charging and dust release mechanisms from surfaces in the PSI experiment; provides estimates of dust charging and release time for a wide range of grain sizes, surface potentials, and plasma conditions relevant to NASA missions using surface and surface/in-depth charging models; delivers an advanced modeling capability that can assist in the design of future ground-based and International Space Station (ISS)-based dusty plasma experiments as well, in the development of active methods for dust removal from extravehicular activity (EVA) suits or instruments exposed to plasmas; and enhances NASA's science readiness with the delivery of a validated capability for dusty plasmas that appear in broader areas of science and engineering.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The research provides testing of hypothesis of dust charging and dust release mechanisms from surfaces in the NASA PSI experiment. It provides estimates of dust charging and release time for a wide range of grain sizes, surface potentials, plasma conditions relevant to NASA missions using surface and surface/in-depth charging models The research delivers an advanced modeling capability that can assist in the design of future ground-based, ISS-based or Lunar-based dust experiments as well in the development of active methods for dust removal from EVA suits or instruments exposed to plasmas. Overall, the research enhances NASA's science readiness with the delivery of a validated computational capability for dusty plasmas that appear in broader areas of science and engineering.
Task Progress:	Plasmas with solid dust particulates (or grains), much more massive than the ions present, are referred to as dusty plasmas and are found in ambient space environments, in the vicinity of spacecraft surfaces, as well as in laboratory and industrial plasma processes. Dusty plasmas in the space environment are influencing the formation of structures surrounding comets, planetary rings, and Earth's upper atmosphere. Dusty plasmas are also formed in the vicinity of spacecraft, due to particulates released after having been trapped on surfaces, due to material degradation, and ice resulting from liquid or propellants. Such near-spacecraft dusty plasmas have been a source of spurious signals in optical sensors. Dusty plasmas have also been recognized to possibly impact extravehicular activities (EVA), as well as operation of equipment on the dusty environments of the Moon and Mars. Numerous laboratory and computational studies have been conducted to further understand the dust charging and transport and removal processes. To alleviate particulate contamination of surfaces, passive and active dust mitigation and removal methods have been developed. Passive includes pre-treatment of surfaces to reduce adhesion, usage of shields to reduce deposition rates; or active, which includes fluidal, mechanical, or electrostatic dust removal. To investigate adhesion properties of the NASA Johnson Space Center (JSC)-1 Lunar Simulant dust to EVA suits, and the effectiveness of plasma in removing the dust, ground-based experiments were conducted in a chamber under a NASA Physical Sciences Informatics (PSI) study of Flanagan and Goree, (2006). In these PSI experiments, electrostatic forces were used to remove dust from surfaces such as spacesuits. The dust was generated on a glass sphere immersed in a neutral neon gas and was found to be released when plasma was formed in the chamber due to the injection of an electron beam. The proposed hypothesis was that a dust grain collects a fluctuating charge, and release from the surface occurs when a dust charge fluctuates to a large negative value, such that the electric force exceeds the adhesive force between dust-surface. First, we conducted a fully kinetic numerical investigation of the charging of spherical and irregular dust grains in the orbitalmotion-limited (OML) sheath regime and a stationary experimental plasma environment, utilizing the Dusty Parallel Immersed-Finite-Element Particle-in-Cell (PIFE-PIC-D) framework further developed in this project. The simulations account for surface charging of the dust grains immersed in a stationary plasma environment similar to PSI. PIFE-PIC-D explicitly resolves the geometrical and material properties (permittivity) of each individual dust grain. We studied the electrostatic interactions in dusty plasmas (particularly, charging of dust particulates in the collision-less regime as the PSI experiment). Considering the size of dust grains (order of microns) vs. the size of the glass sphere (order of cm) used in the PSI experiment, the microscopic dust grain charging in the PSI experiment can be modeled as the configurations of dust grains near a "flat" surface, which were simulated using the recently developed PIFE-PIC-D code suite. A unique feature of our study is that both the permittivity and irregular shape of dust grains were explicitly resolved in PIFE-PIC-D. In this study, "irregular" shapes were achieved through patching of spheres. Multiple dust grain configurations were compared to see how dust grains are charged in stationary and drifting plasma environments. These included: single dust grain in a stationary plasma, multiple dust grains in a stationary plasma, multiple irregularly shaped dust grains in a stationary plasma, single/multiple dust grains in a drifting plasma, and single/multiple dust grains in a drifting plasma near a surface. The charge collection over time of each dust grain was investigated with varying size, irregularity, number of grains, spacing between dust grains, and permittivity. A notable product of this project is the highly parallel PIFE-PIC-D code suite, and its capability to incorporate the dielectric permittivity of the grain material, allowing computation of the potential distribution over the grain's surface. This novel aspect extends our understanding of charging phenomena for both spherical and irregular dust grains. In contrast to prior studies that focused on fully conducting or perfectly dielectric spheres, this work explored a broader range of permittivities for irregular dust grain aggregates. The charging behavior of a dust cluster was estimated by calculating its electron Debye length edge-to-edge separation to offer valuable insights into a dust cluster's general charge dynamics. Furthermore, this work introduced the first fully kinetic simulations that vary the permittivity of the grain material, while resolving an arbitrary dust grain surface used to create an irregular grain immersed in a stationary experimental plasma environment.

The findings emphasize the grain's permittivity not only influences the overall net surface charge ratio, but also its surface potential distribution. While estimating the charging behavior of irregular grains, using the spherical grain case as a reference, is common, our findings emphasize that this approach may underestimate results. Thus, achieving accurate results demands explicit consideration of the grain's geometry and permittivity, especially in scenarios where non-uniform charging dynamics will be prominent.

Second, we conducted a fully kinetic investigation of the macroscopic lunar surface charging as one of the drivers of dust grain charging and transport, which motivated the PSI experiment. These simulations were performed with the PIFE-PIC code, which can resolve the uneven lunar surface terrain as well as landers and habitats on the surface of the Moon. Particularly, this research considered the plasma charging near the lunar surface for future exploration missions, specifically, near lunar craters at the terminator region which are target destinations for planned Artemis missions. Under ambient solar wind and photoemission plasma conditions, the rugged surface terrain could generate localized plasma wakes and shadow regions, which can lead to strong differential charging of the surface. Such localized plasma flow field, together with the charged lunar surface, provides an electric field leading to lofting and levitation of charged dust grains. The simulations provided predictions of that regolith surface charging around a small crater at the lunar terminator. The results clearly show the differential charging of the lunar surface and strengthens the expectations that the thermal electron temperature plays a role in altering the surface potential. Such macroscopic PIFE-PIC simulations will provide boundary conditions needed for microscopic simulation of dust charging, lofting and levitation, near surfaces on the Moon.

Third, we investigated numerically, using PIC simulations, the process of dust charging and release from dielectric surfaces under conditions found in the PSI dust removal experiments. We sought to contribute to the understanding and prediction for dielectric dust removal from a conductive or a dielectric surface by plasma or electron beams. The Spacecraft Plasma Interaction System (SPIS) software was employed to simulate the multiscale charging at both the grain surface scale (μm) and system-scale. Such PIC simulations require a multi-grid approach to capture charging effects at the microscopic (μm), and macroscopic scales (mm to cm), and to model the self-consistent charging process of dust particles and the surface. Meshes are generated by Gmsh and used with SPIS. [Ed. Note: Gmsh is a 3D finite element mesh generator.] We performed simulations of an isolated conductive dust charging in a thermal plasma to verify and benchmark SPIS. These simulations were compared with current-collection theory covering the thin-sheath to OML regimes for a variety of applied potentials. We also conducted simulations of an isolated dielectric dust under conditions found in the PSI experiments that include thermal plasmas, an electron beam, and thermal plasmas with an electron beam. Findings from the simulations of an isolated dielectric spherical dust, using assumed PSI experiment conditions, showed good agreement with OML theory. Simulations of a single dielectric dust particle on the surface of a larger dielectric sphere maintained similar charging characteristics to isolated results, but also provided surface electric fields between the dust and the larger sphere. Force calculations made considering the effect of this electric field and the charge on the dust were compared to estimates of the adhesive force, allowing prediction on whether dust release was possible. The force calculations suggested, in agreement with the PSI, that beam electrons alone were not sufficient to release the dust, and that the combined case with both thermal plasma and a beam was capable of release. The results showed that the detailed plasma and beam properties, as well as the dielectric properties of the dust particles of the PSI or any other experiment, are necessary to make accurate predictions. The use of dust layers is also essential to make further and direct comparisons with the PSI experiment.

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